R&D PROJECT

# Formal R&D project proposal / statement of work

Prepared for NRC IRAP discussion | 2026-07-12

## Project objective

Opyjo Inc. will determine whether a privacy-preserving, content-agnostic adaptive engine can infer reliable learner state from sparse outcomes and select practice that improves durable learning across tenants.

## Technical uncertainties

The project will test sparse-data mastery estimation, correction of item-difficulty priors from behaviour, durable outcome improvement over a fixed baseline, and safe generalization across different curricula and traffic patterns.

## Systematic work

WP1 compares EWMA, Elo/IRT and confidence-weighted learner models. WP2 develops empirical item calibration and drift controls. WP3 conducts a stable adaptive-versus-baseline outcome experiment. WP4 validates isolation, latency, coverage and mapping with external tenants.

## Milestones and deliverables

M1 model-selection report and versioned parameters (Month 3); M2 calibration pipeline and review thresholds (Month 5); M3 controlled outcome report (Month 8); M4 external-tenant validation and onboarding playbook (Month 10).

## Success measures

Calibration error, log-loss, cold-start accuracy, prior-versus-observed difficulty error, delayed retention, time-to-mastery, recommendation coverage, p95 latency, tenant-isolation tests, and reproducible model decisions.

## Starting point and advancement sought

VERIFIED FROM REPOSITORY (July 12, 2026)
Product: standalone, content-agnostic, multi-tenant adaptive-learning API.
Runtime: Node.js 24, TypeScript 5.9, Fastify 5, PostgreSQL 17 locally; Docker image and Railway deployment configuration are present.
Validation: 16 automated tests pass; 3 database integration tests are defined but skipped without TEST_DATABASE_URL. TypeScript build and ESLint pass.
Current status: code-complete and locally validated. Production deployment, managed-database migration execution, restore/load/privacy testing, Brightwick shadow integration, and learning-outcome validation must not be represented as complete until evidence exists.

The baseline adaptive-v1 model is a difficulty-weighted EWMA beginning at 0.5 with learning rate 0.15. It scaffolds after two consecutive wrong answers, targets the weakest skill below 0.8, otherwise spirals the least recently practised skill, maps mastery to difficulty bands, and avoids the last 30 items or 14 days where possible. This is an engineered hypothesis—not yet evidence of accurate learner inference or durable learning gains.

Advancement sought: a confidence-aware, empirically calibrated, privacy-preserving learner-and-item decision system that remains reliable with sparse data and heterogeneous tenant curricula, supported by reproducible outcome evidence.

## WP1 - Sparse-data learner model

Question: can learner state be estimated reliably before enough responses exist for conventional calibration? Compare adaptive-v1 EWMA against Elo/IRT-inspired and confidence-weighted alternatives using replay simulation and, when available, prospective data. Define train/validation separation, cold-start cohorts, missing-data handling, versioned parameters, and rollback. Acceptance evidence: log-loss/Brier score, calibration error, ranking stability, uncertainty coverage, and observations-to-stability by cohort. Deliverable: model-selection report, code, parameter registry, limitations, and decision.

## WP2 - Empirical item difficulty

Question: can tenant-supplied 1-5 difficulty priors be corrected without overreacting to biased or low-volume behaviour? Estimate performance conditional on learner state; define minimum samples, confidence intervals, shrinkage, drift monitoring, tenant/global pooling rules, and human-review thresholds. Acceptance evidence: prior-versus-observed error, held-out predictive improvement, recalibration stability, drift false-positive rate, and documented failure cases. Deliverable: versioned calibration pipeline and audit report.

## WP3 - Durable outcome experiment

Question: does adaptive-v1 or its successor improve retained learning rather than only immediate correctness or engagement? Use stable learner assignment and an agreed baseline; pre-register primary outcome, delayed-review window, exclusions, sample-size rationale, stopping rules, guardrails, and analysis. Primary candidates: delayed retention and time-to-mastery. Guardrails: completion, error streak, fallback rate, latency, support incidents, and subgroup harm. Deliverable: reproducible analysis and promote/revise/rollback decision.

## WP4 - Cross-tenant validation

Question: does the content-agnostic model remain useful and safe across different skill taxonomies, banks, volumes, and integration patterns? Onboard 3-5 design partners using metadata-only contracts and opaque IDs. Test mapping quality, recommendation coverage, tenant isolation, retry/idempotency, latency, outage fallback, deletion/export procedures, and integration time. Deliverable: partner evidence, hardened onboarding playbook, limitations, and commercial-readiness decision.

## Milestones, governance, and Canadian benefit

M1 Month 1: experimental protocol and baseline evidence pack. M2 Month 3: WP1 decision. M3 Month 5: WP2 calibrated pipeline. M4 Month 8: WP3 outcome report. M5 Month 10: WP4 validation and commercialization gate. Monthly technical reviews retain hypotheses, commits, datasets, results, failures, timesheets, and decisions. Expected Canadian benefit must be completed with verified numbers: [INPUT] Canadian R&D jobs retained/created, [INPUT] revenue/export potential, Canadian-owned IP position, and new in-house modelling/security capability.

## Submission completion

SUBMISSION INPUTS STILL REQUIRED

[INPUT] Legal corporation name, incorporation jurisdiction/date, CRA business number, registered address, fiscal year-end, and ownership percentages.

[INPUT] Project start/end dates and the ITA-approved scope.

[INPUT] Named project personnel, Canadian work location, employment status, salary, productive-hours method, and monthly project hours.

[INPUT] Actual historical financial statements, bank balances, financing commitments, tax balances, and monthly forecast assumptions.

[INPUT] Current customers/revenue, interview evidence, pipeline, pricing validation, market-source citations, and signed partner evidence.

[INPUT] Requested contribution, confirmed eligible-cost treatment, claim frequency, and other government assistance.

These fields must be completed from source records. The application must not substitute estimates for verified facts without labelling them as assumptions.